

Chapter 8

Big Data Analytics in the Cloud

Cloud Computing Solutions Architect

A Hands-On Approach

A Competency-based Textbook for Universities and
a Guide for AWS Cloud Certification and Beyond



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Outline

- What is Analytics?
- What is Big Data?
- Characteristics of Big Data
- Domain Specific Examples of Big Data
- Analytics Flow for Big Data
- Big Data Stack
- Mapping Analytics Flow to Big Data Stack
- Analytics Patterns

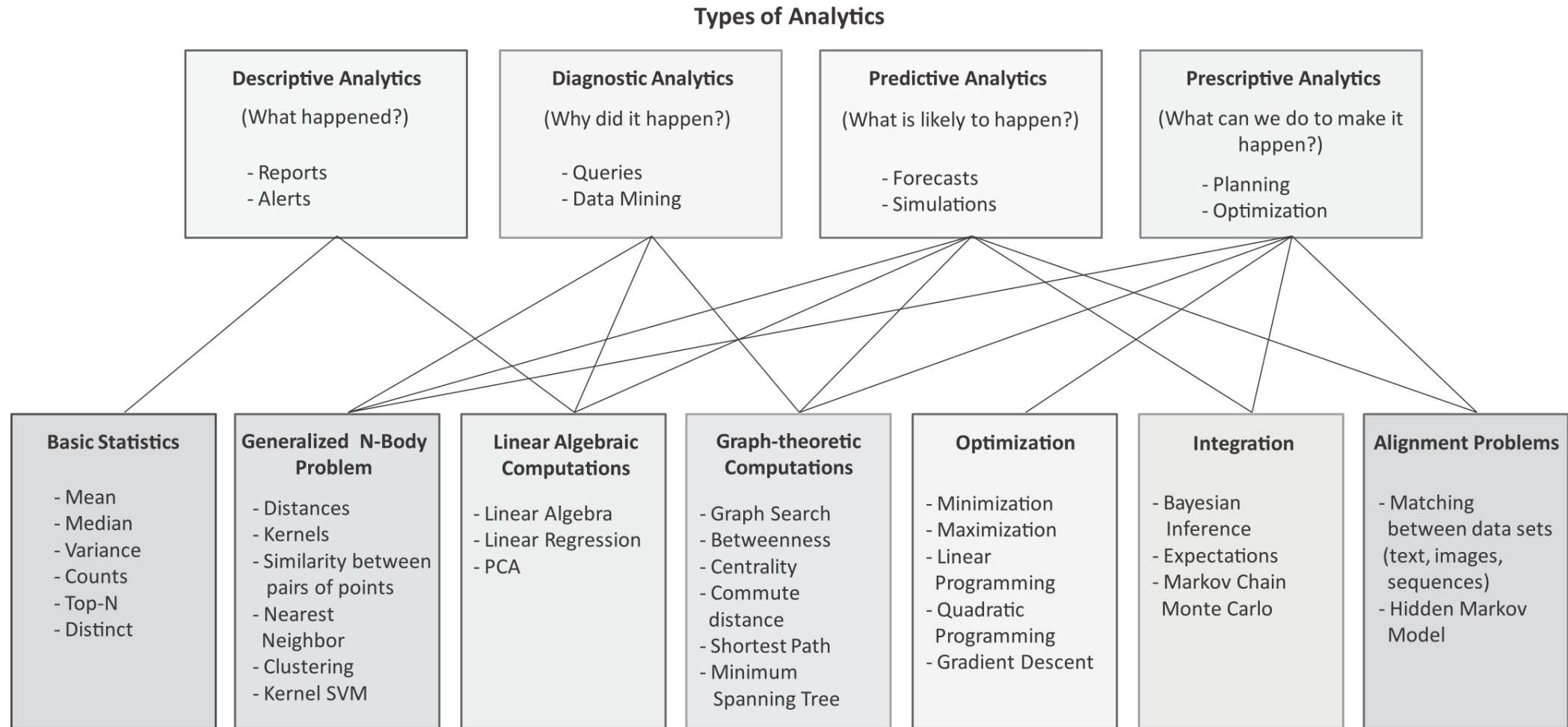
What is Analytics?

- Analytics is a broad term that encompasses the processes, technologies, frameworks and algorithms to extract meaningful insights from data.
- Raw data in itself does not have a meaning until it is contextualized and processed into useful information.
- Analytics is this process of extracting and creating information from raw data by filtering, processing, categorizing, condensing and contextualizing the data.
- This information obtained is then organized and structured to infer knowledge about the system and/or its users, its environment, and its operations and progress towards its objectives, thus making the systems smarter and more efficient.

The Seven Giants

- The National Research Council has done a characterization of computational tasks for massive data analysis (called the seven “giants”).
- These computational tasks include:
 - (1) Basis Statistics,
 - (2) Generalized N-Body Problems,
 - (3) Linear Algebraic Computations,
 - (4) Graph-Theoretic Computations,
 - (5) Optimization,
 - (6) Integration and
 - (7) Alignment Problems.
- This characterization of computational tasks aims to provide a taxonomy of tasks that have proved to be useful in data analysis and grouping them roughly according to mathematical structure and computational strategy.

Mapping between types of analytics and computational tasks or 'giants'



Types of Analytics

- **Descriptive Analytics**

- Descriptive analytics comprises analyzing past data to present it in a summarized form which can be easily interpreted. Descriptive analytics aims to answer - ***What has happened?*** A major portion of analytics done today is descriptive analytics through use of statistics functions such as counts, maximum, minimum, mean, top-N, percentage, for instance. These statistics help in describing patterns in the data and present the data in a summarized form.

- **Diagnostic Analytics**

- Diagnostic analytics comprises analysis of past data to diagnose the reasons as to why certain events happened. Diagnostic analytics aims to answer - ***Why did it happen?*** While descriptive analytics can be useful for summarizing the data by computing various statistics (such as mean, minimum, maximum, variance, or top-N), diagnostic analytics can provide more insights into why certain a fault has occurred based on the patterns in the sensor data for previous faults.

Types of Analytics

- **Predictive Analytics**

- Predictive analytics comprises predicting the occurrence of an event or the likely outcome of an event or forecasting the future values using prediction models. Predictive analytics aims to answer - ***What is likely to happen?*** Predictive Analytics is done using predictive models which are trained by existing data. These models learn patterns and trends from the existing data and predict the occurrence of an event or the likely outcome of an event (classification models) or forecast numbers (regression models).

- **Prescriptive Analytics**

- While predictive analytics uses prediction models to predict the likely outcome of an event, prescriptive analytics uses multiple prediction models to predict various outcomes and the best course of action for each outcome. Prescriptive analytics aims to answer - ***What can we do to make it happen?*** Prescriptive Analytics can predict the possible outcomes based on the current choice of actions.

What is Big Data?

- Big data is defined as collections of datasets whose volume, velocity or variety is so large that it is difficult to store, manage, process and analyze the data using traditional databases and data processing tools.
- According to an estimate by IBM, 2.5 quintillion bytes of data is created every day.
- A recent report by DOMO estimates the amount of data generated every minute on popular online platforms. Below are some key pieces of data from the report:
 - Facebook users share nearly 4.16 million pieces of content
 - Twitter users send nearly 300,000 tweets
 - Instagram users like nearly 1.73 million photos
 - YouTube users upload 300 hours of new video content
 - Apple users download nearly 51,000 apps
 - Skype users make nearly 110,000 new calls
 - Amazon receives 4300 new visitors
 - Uber passengers take 694 rides
 - Netflix subscribers stream nearly 77,000 hours of video

Big Data Analytics

- Big Data analytics deals with collection, storage, processing and analysis of this **massive-scale data**.
- Big data analytics involves several steps starting from data cleansing, data munging (or wrangling), data processing and visualization.
- Specialized tools and frameworks are required for big data analysis when:
 - the volume of data involved is so large that it is difficult to store, process and analyze data on a single machine,
 - the velocity of data is very high and the data needs to be analyzed in real-time,
 - there is variety of data involved, which can be structured, unstructured or semi-structured, and is collected from multiple data sources,
 - various types of analytics need to be performed to extract value from the data such as descriptive, diagnostic, predictive and prescriptive analytics.

Big Data Examples

- Data generated by social networks including text, images, audio and video data
- Click-stream data generated by web applications such as e-Commerce to analyze user behavior
- Machine sensor data collected from sensors embedded in industrial and energy systems for monitoring their health and detecting failures
- Healthcare data collected in electronic health record (EHR) systems
- Logs generated by web applications
- Stock markets data
- Transactional data generated by banking and financial applications

Characteristics of Big Data

- Volume
 - Big data is a form of data whose volume is so large that it would not fit on a single machine therefore specialized tools and frameworks are required to store process and analyze such data.
- Velocity
 - Velocity of data refers to how fast the data is generated. Data generated by certain sources can arrive at very high velocities, for example, social media data or sensor data.
- Variety
 - Variety refers to the forms of the data. Big data comes in different forms such as structured, unstructured or semi-structured, including text data, image, audio, video and sensor data.

Characteristics of Big Data

- Veracity
 - Veracity refers to how accurate is the data. To extract value from the data, the data needs to be cleaned to remove noise.
- Value
 - Value of data refers to the usefulness of data for the intended purpose. The end goal of any big data analytics system is to extract value from the data.

Domain Specific Examples of Big Data

- Web
- Financial
- Healthcare
- Internet of Things
- Environment
- Logistics & Transportation
- Industry
- Retail

Analytics Flow for Big Data

- **Data Collection**

- Data collection is the first step for any analytics application. Before the data can be analyzed, the data must be collected and ingested into a big data stack. The choice of tools and frameworks for data collection depends on the source of data and the type of data being ingested.

- **Data Preparation**

- Data can often be dirty and can have various issues that must be resolved before the data can be processed, such as corrupt records, missing values, duplicates, inconsistent abbreviations, inconsistent units, typos, incorrect spellings and incorrect formatting. Data preparation step involves various tasks such as data cleansing, data wrangling or munging, de-duplication, normalization, sampling and filtering.

Analytics Flow for Big Data

- **Analysis Types**

- The next step in the analysis flow is to determine the analysis type for the application.

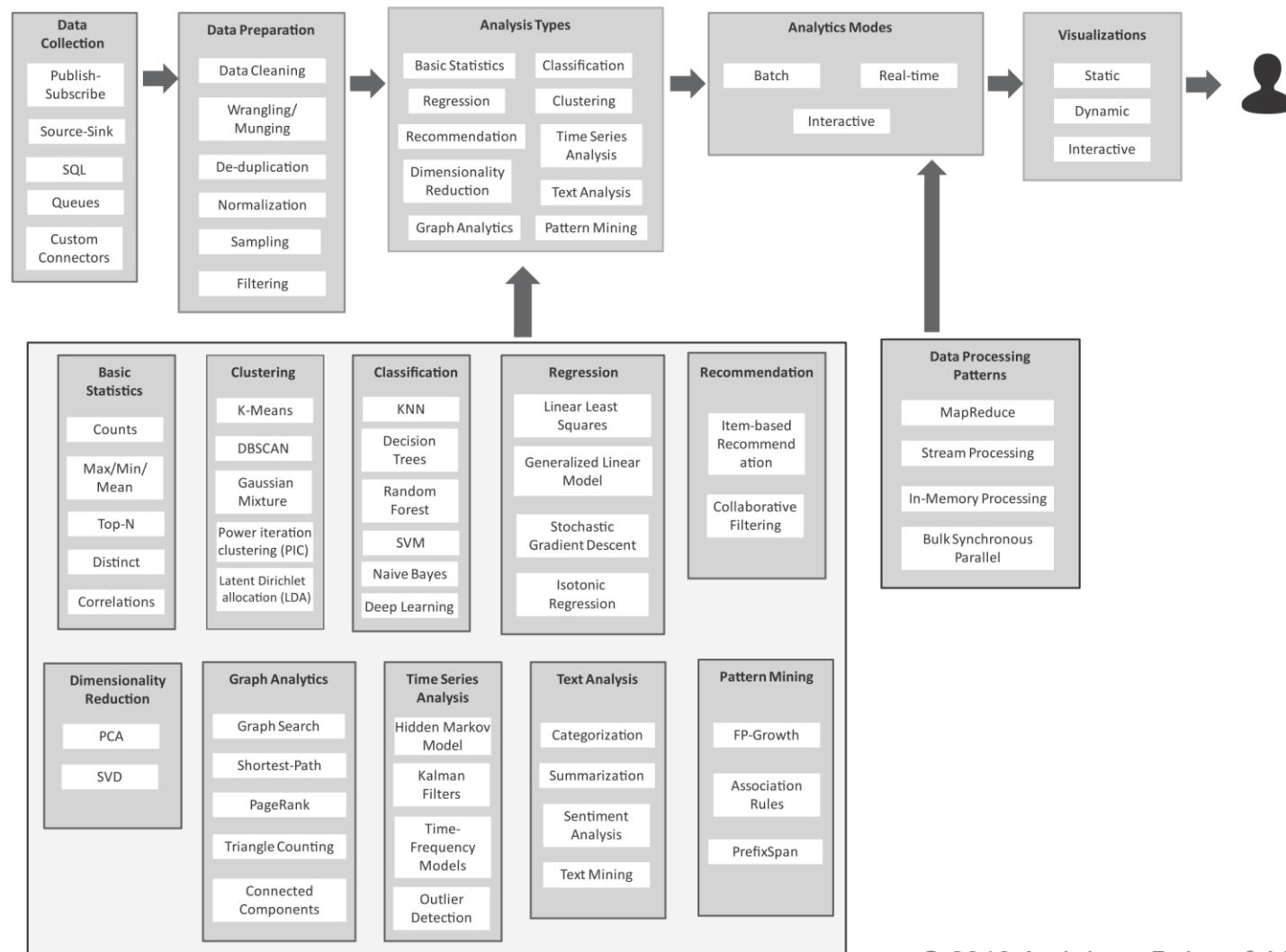
- **Analysis Modes**

- With the analysis types selected for an application, the next step is to determine the analysis mode, which can be either batch, real-time or interactive. The choice of the mode depends on the requirements of the application.

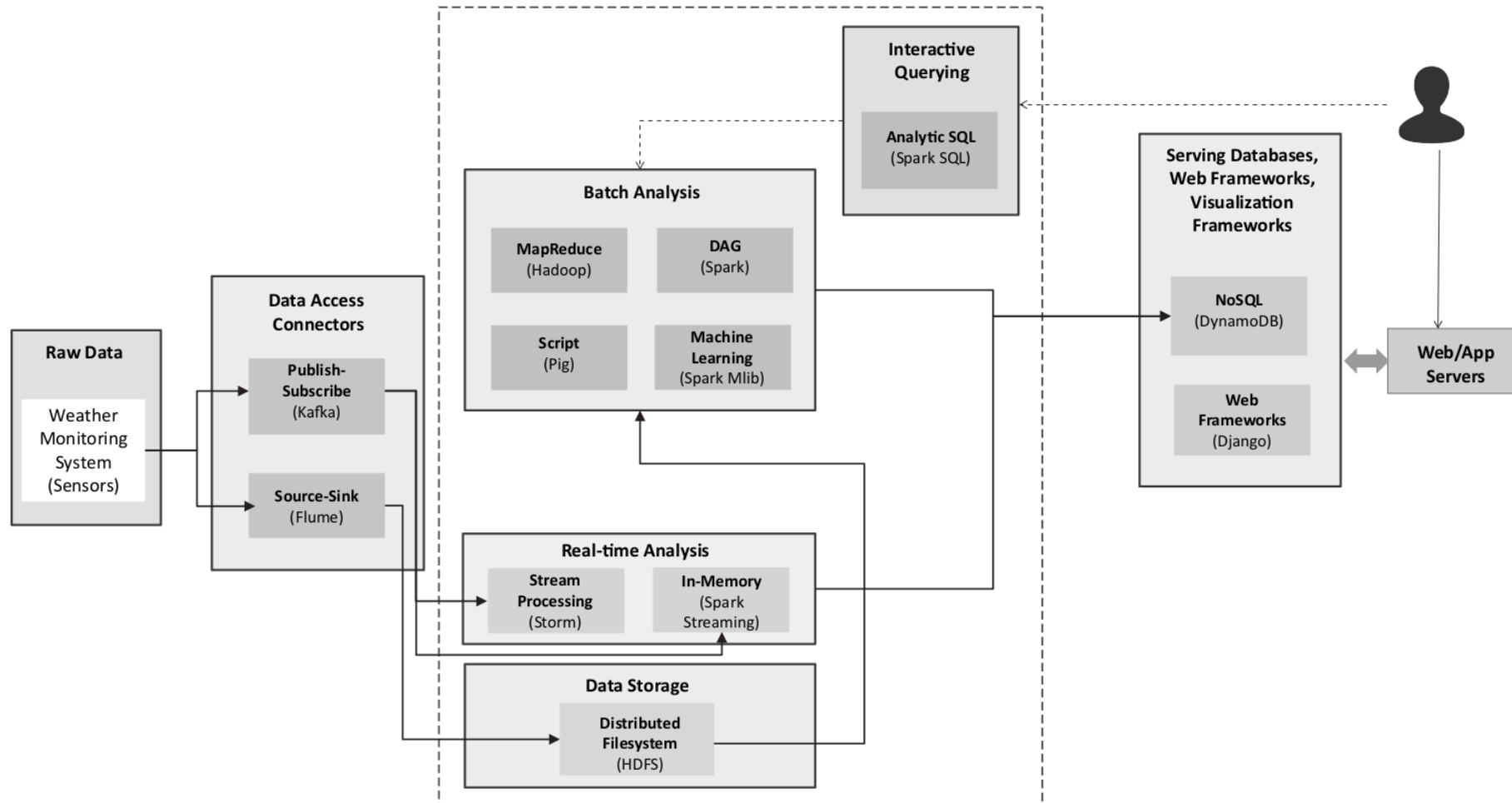
- **Visualizations**

- The choice of the visualization tools, serving databases and web frameworks is driven by the requirements of the application. Visualizations can be static, dynamic or interactive.

Analytics Flow for Big Data



Big Data Stack



Big Data Stack

- **Raw Data Sources**

- In any big data analytics application or platform, before the data is processed and analyzed, it must be captured from the raw data sources into the big data systems and frameworks.

- **Data Access Connectors**

- The Data Access Connectors includes tools and frameworks for collecting and ingesting data from various sources into the big data storage and analytics frameworks. **The choice of the data connector is driven by the type of the data source.**

- **Data Storage**

- The data storage block in the big data stack includes distributed filesystems and non-relational (NoSQL) databases, which store the data collected from the raw data sources using the data access connectors.

Big Data Stack

- **Batch Analytics**
 - The batch analytics block in the big data stack includes various frameworks which allow analysis of data in batches such as Hadoop-MapReduce, Pig, Oozie, Pig, Spark, Solr.
- **Real-time Analytics**
 - The real-time analytics block includes the Apache Storm and Spark Streaming frameworks.
- **Interactive Querying**
 - Interactive querying systems allow users to query data by writing statements in SQL-like languages.
- **Serving Databases, Web & Visualization Frameworks**
 - **While the various analytics blocks process and analyze the data, the results are stored in serving databases for subsequent tasks of presentation and visualization. These serving databases allow the analyzed data to be queried and presented in the web applications.**

Mapping Analytics Flow to Big Data Stack

Data Collection

Analysis Type	Framework (Mode)
Publish-Subscribe	Kafka, Kinesis
Source-Sink	Flume
SQL	Sqoop
Queues	SQS, RabbitMQ, ZeroMQ, RESTMQ
Custom Connectors	REST, WebSocket, MQTT

Data Preparation

Analysis Type	Framework
Data Cleaning	Open Refine
Data Wrangling	Open Refine DataWrangler
De-Duplication	Open Refine, Pig, Hive, Spark SQL
Normalization Sampling, Filtering	MapReduce, Pig, Hive, Spark SQL

Basic Statistics

Analysis Type	Framework (Mode)
Counts, Max, Min, Mean, Top-N, Distinct	Hadoop-MapReduce (Batch), Pig (Batch), Spark (Batch), Spark Streaming (Realtime), Spark SQL (Interactive), Hive (Integrative), Storm (Real-time)
Correlations	Hadoop-MapReduce (Batch), Spark Mlib (Batch)

Clustering

Analysis Type	Framework (Mode)
K-Means	Hadoop-MapReduce (Batch), Spark Mlib (Batch & Real-time) H2O (Batch)
DBSCAN	Spark (Batch)
Gaussian Mixture	Spark Mlib (Batch)
PIC	Spark Mlib (Batch)
LDA	Spark Mlib (Batch)

Classification

Analysis Type	Framework (Mode)
KNN	Spark Mlib (Batch , Realtime)
Decision Trees	Spark Mlib (Batch, Realtime)
Random Forest	Spark Mlib (Batch , Realtime), H2O (Batch)
SVM	Spark Mlib (Batch , Realtime)
Naïve Bayes	Spark Mlib (Batch, Realtime), H2O (Batch)
Deep Learning	H2O (Batch)

Regression

Analysis Type	Framework (Mode)
Linear Least Squares	Spark Mlib (Batch, Realtime)
Generalized Linear Model	H2O (Batch)
Stochastic Gradient Descent	Spark Mlib (Batch, Realtime)
Isotonic Regression	Spark Mlib (Batch, Realtime)

Mapping Analytics Flow to Big Data Stack

Graph Analytics

Analysis Type	Framework (Mode)
Graph Search	Spark GraphX (Batch)
Shortest-Path	Spark GraphX (Batch)
PageRank	Spark GraphX (Batch)
Triangle Counting	Spark GraphX (Batch)
Connected Components	Spark GraphX (Batch)

Time Series Analysis

Analysis Type	Framework (Mode)
Kalman Filter	Spark (Realtime)
Time Frequency Models	Spark (Realtime)
Outlier Detection	Storm (Realtime), Spark (Batch, Realtime)

Dimensionality Reduction

Analysis Type	Framework (Mode)
SVD	Spark Mlib (Batch)
PCA	Spark Mlib (Batch), H2O (Batch)

Recommendation

Analysis Type	Framework (Mode)
Item-bases Recommendation	Spark Mlib (Batch)
Collaborative Filtering	Spark Mlib (Batch)

Text Analysis

Analysis Type	Framework (Mode)
Categorization	Hadoop-MapReduce (Batch), Storm (Realtime), Spark (Batch, Realtime)
Summarization	Spark (Batch)
Sentiment Analysis	Storm (Realtime), Spark (Batch, Realtime)
Text Mining	Storm (Realtime), Spark (Batch, Realtime)

Pattern Mining

Analysis Type	Framework (Mode)
FP-Growth	Spark Mlib (Batch)
Association Rules	Spark Mlib (Batch)
PrefixSpan	Spark Mlib (Batch)

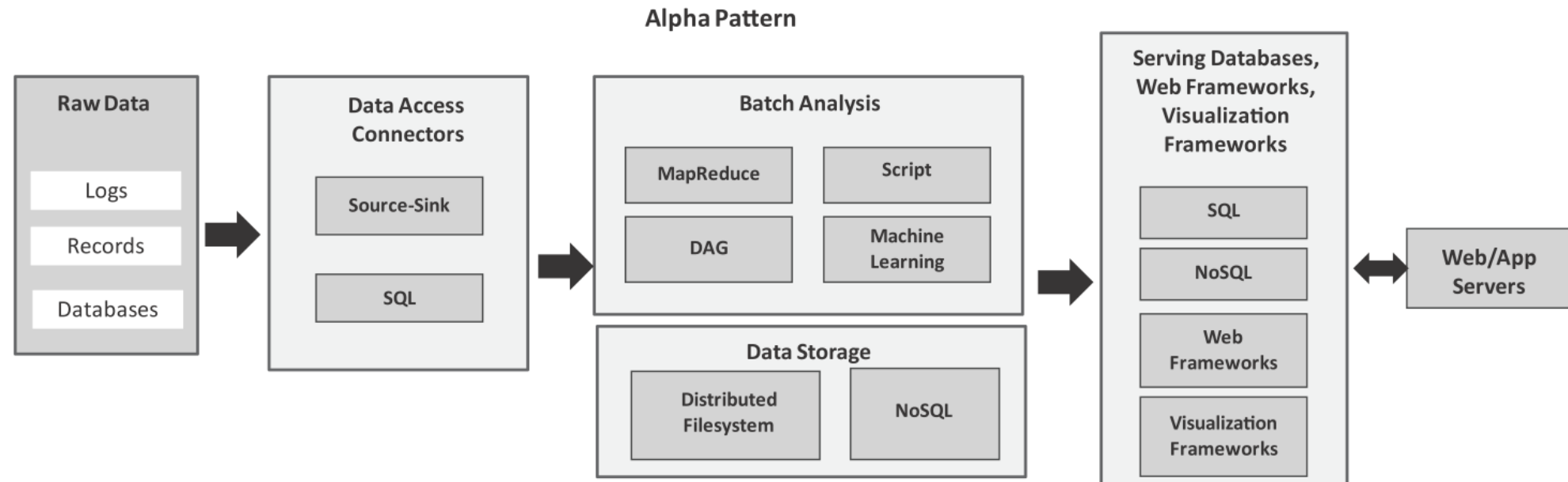
Visualization

Analysis Type	Framework (Mode)
Web Frameworks	Django, Flask
SQL Databases	MySQL
NoSQL Databases	Hbase, DynamoDB, Cassandra, MongoDB
Visualization Frameworks	Lightning, pyGal, Seaborn

Analytics Patterns

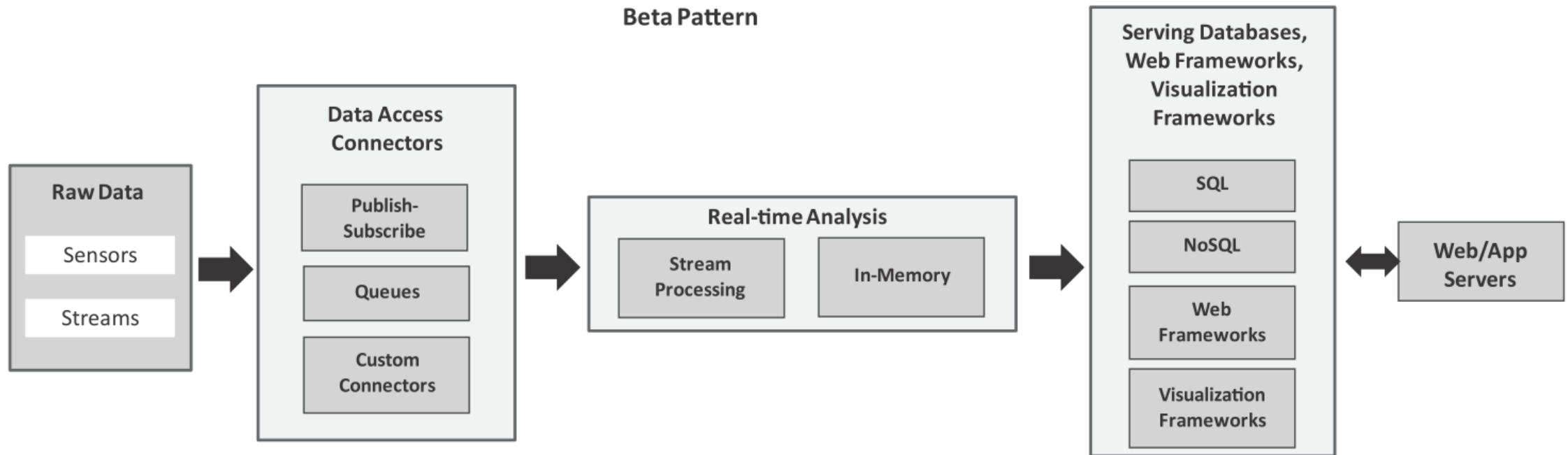
- **Alpha Pattern**

- This pattern can be used for ingesting large volumes of data into a distributed filesystem (such as HDFS) or a NoSQL database (such as HBase) using source-sink connectors (such as Flume) and SQL connectors (such as Sqoop). After the data is moved to the stack, the data can be analyzed in batch mode with batch analysis frameworks including MapReduce (using Hadoop), scripting frameworks (such as Pig), distributed acyclic graph frameworks (such as Spark), machine learning frameworks (such as Spark MLlib). The analysis results are stored either in relational or non-relational databases.



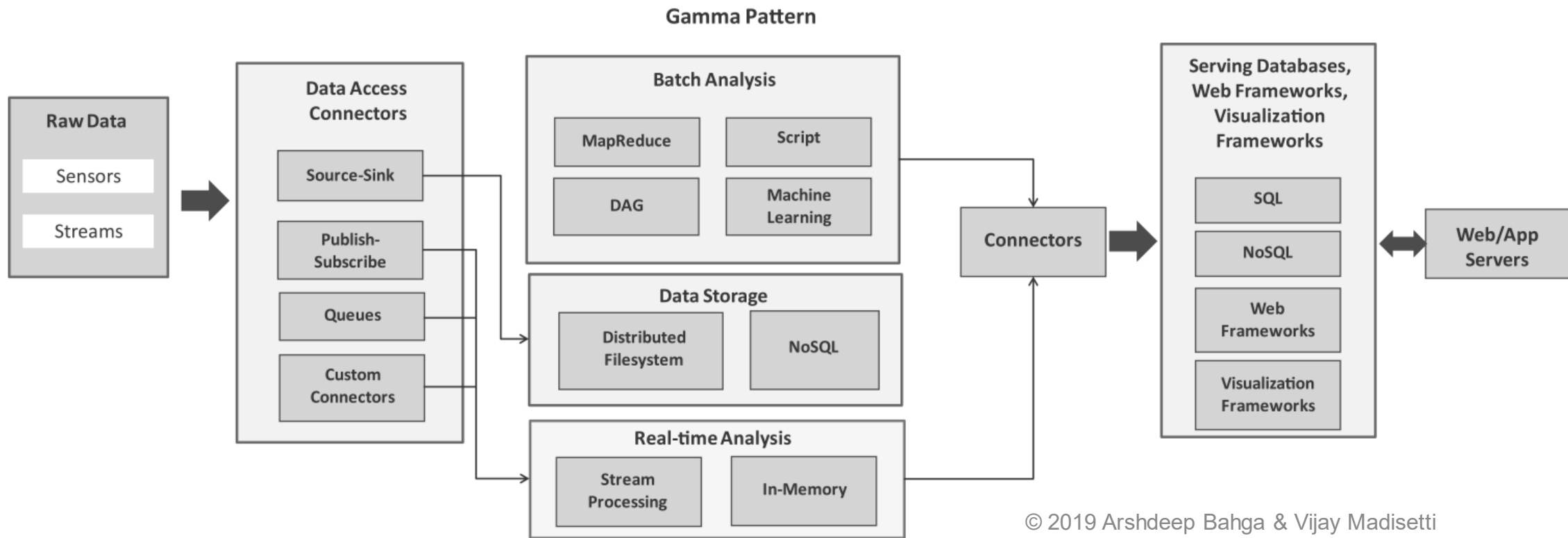
Analytics Patterns

- **Beta Pattern:**
 - This pattern can be used for ingesting streaming data using publish-subscribe messaging frameworks, queues and custom connectors. For real-time analysis, we can use stream processing frameworks (such as Storm) or in-memory processing frameworks (such as Spark).



Analytics Patterns

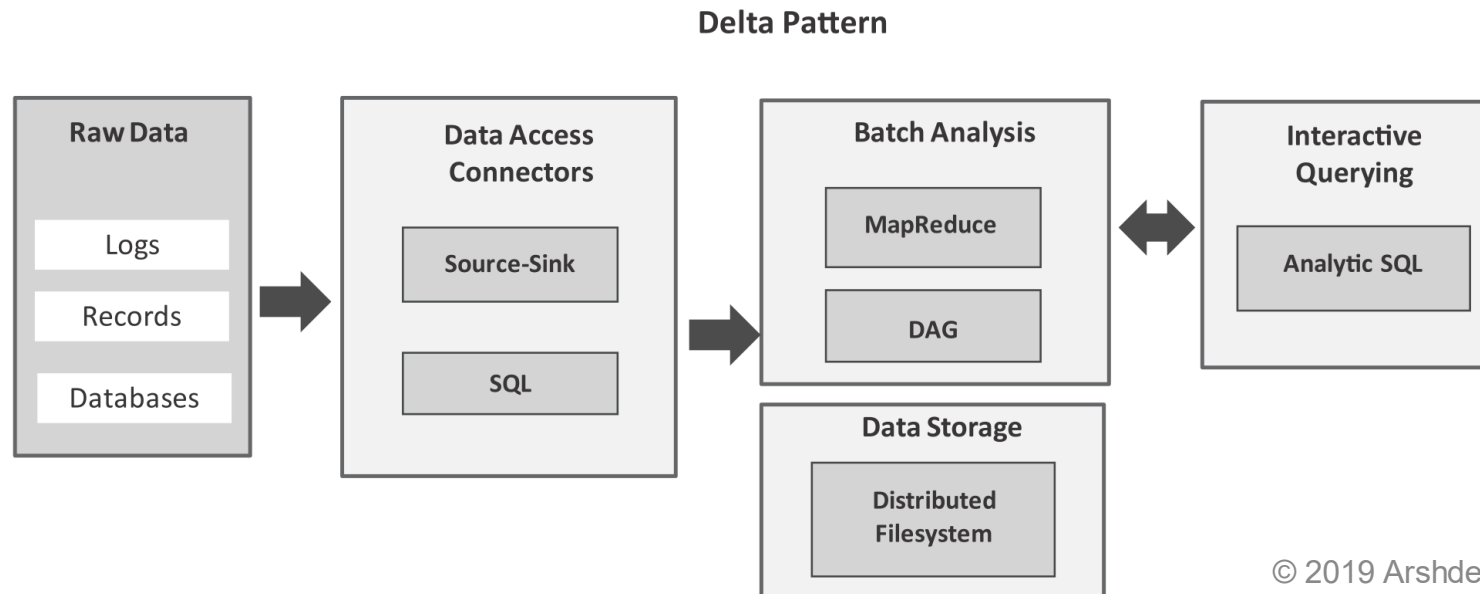
- **Gamma Pattern:**
 - This pattern is meant for ingesting streaming data into a big data stack and analyzing the data both in real-time and in batch modes.



Analytics Patterns

- **Delta Pattern:**

- This pattern uses, source-sink connectors (such as Flume) or SQL connectors (such as Sqoop) to ingest bulk data into the big data stack. After the data is moved to a distributed filesystem, you can use interactive querying frameworks (such as Hive or Spark SQL) for querying data with SQL-like queries in an interactive mode.



Mapping of blocks in analytics patterns to AWS services

Data Access Connector	AWS Service
Publish-Subscribe	AWS Kinesis
Source-Sink	Flume on EMR
SQL	Sqoop on EMR
Queues	AWS SQS
Custom Connectors	AWS IoT

Data Storage	AWS Service
Distributed Filesystem	HDFS on AWS EMR
NoSQL	AWS DynamoDB

Batch Analysis	AWS Service
MapReduce	Hadoop on AWS EMR
Script	Pig on AWS EMR
DAG	Spark on AWS EMR
Machine Learning	Spark MLlib on AWS EMR, AWS Machine Learning

Real-time Analysis	AWS Service
Stream Processing	Storm cluster on AWS EC2
In-Memory Processing	Spark on AWS EMR

Serving Databases, Web & Visualization Frameworks	AWS Service
SQL	AWS RDS
NoSQL	AWS DynamoDB
Web Framework	Django on AWS EC2 instance
Visualization Framework	Lightning, Pygal, Seaborn, on AWS Virtual Machines instance

Complexity Levels for Analytics Patterns

Levels	Security	• Apache Ranger: Authorization, authentication, auditing, data encryption, security administration • Apache Knox Gateway: REST API gateway, LDAP and Active Directory Authentication, Federation/SSO, Authorization, Auditing			
	Fault Tolerance	• Distributed, fault tolerant and highly available architectures • Data replication, automatic failovers			
	Scalability	Scalable to thousands of nodes that can store and process several Petabytes of data	Scalable to thousands of nodes that can process very high throughput streaming data	Scalable to thousands of nodes that can process very high throughput streaming data	Scalable to thousands of nodes that can store and process several Petabytes of data
	Performance	Process large volumes of data with within timescales of few minutes	Process streaming data on the timescales of few milliseconds to seconds	Combines batch and real-time processing for streaming data, with timescales of milliseconds to seconds for real-time, and few minutes for batch	Interactively query large volumes of data with within timescales of few milliseconds to seconds
	Functionality	Batch Processing	Real time processing	Batch & Real-time processing	Interactive querying
		Alpha	Beta	Gamma	Delta
Analytics Patterns					